

Vedant Ankush Karande

New Delhi, India | +91 70216-29146 | vedantk6403@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Current M.Tech Cybersecurity student at IIT Delhi with a strong foundation in Network Security, Cryptography, and AI integration. GATE CSE 2025 qualified (AIR 1197). Passionate about developing secure systems, demonstrated through patented research in AI-based healthcare diagnostics, advanced cryptographic implementations, and low-level exploit development.

EDUCATION

Indian Institute of Technology (IIT), Delhi <i>M.Tech in Cybersecurity</i> – Relevant Coursework: Cryptography, Network and System Security (SIL7165), System Concepts, High Performance Computers.	New Delhi, India <i>Jul 2025 – Present</i>
Shivaji University <i>B.Tech in Computer Science and Technology</i>	Kolhapur, India <i>2021 – 2025</i>

TECHNICAL SKILLS

Security Tools: Nmap, Burp Suite, Kali Linux, Wireshark, GDB, Pwntools, TryHackMe
Languages: Python, C, Assembly (x86_64), C++ (HackerRank 4-Star), JavaScript, SQL
Frameworks: Flask, TensorFlow, Bootstrap
Core Concepts: Network Security, Low-Level Exploitation, Cryptanalysis, Data Structures

ACHIEVEMENT

GATE CSE 2025: Secured All India Rank (AIR) **1197**.

PUBLICATIONS & PATENTS

Patent: Published a patent titled "Artificial Intelligence based Blood Testing Device" (January 2025).
Conference Paper: Presented a paper on "AI Chatbot for Dermatologists" at the National Conference on Interdisciplinary Advances in Sustainability & Technology (CIAST_25), organized by the Department of Technology, Shivaji University, Kolhapur (March 2025).

PROJECTS

- Bypassing 64-bit strcpy Constraints: A Partial Overwrite Stack Overflow | C, GDB, Pwntools**
- Constructed a vulnerable-by-design x86_64 Linux environment to research memory corruption and overcome the "Null Byte Wall" inherent to 64-bit string-copying functions.
 - Executed a surgically precise partial overwrite attack to hijack the Instruction Pointer (RIP) and bypass `strcpy` null-byte constraints.
 - Demonstrated advanced low-level exploitation by manipulating the SUID bit to escalate privileges and secure a root shell.
- Cryptographic Implementations | M.Tech Coursework**
- Developed scripts to perform cryptanalysis, successfully automating attacks to break **Vigenère** and **Substitution** ciphers.
 - Implemented algorithmic solutions for the **Discrete Logarithm Problem (DLP)** as part of advanced cryptography studies.
- AI Chatbot for Dermatologists | Python, Flask, TensorFlow**
- Developed an AI-powered diagnostic support tool to assist dermatologists by collecting preliminary patient history.
 - Built a Python backend using **Flask** to create an API that communicated with the **TensorFlow** model.

POSITIONS OF RESPONSIBILITY

ISTE Student Chapter (College Committee) <i>Technical Head</i> – Managed website development, version control, and hosting for college events.	Kolhapur, India <i>2023 – 2025</i>
GeeksforGeeks Student Chapter <i>Co-Technical Head</i> – Led technical workshops and lectures for peer learning and managed event logistics.	Kolhapur, India <i>2023 – 2025</i>
National Service Scheme (NSS) <i>NSS Leader</i> – Led a team of volunteers in event management; Awarded "Best NSS Volunteer".	Mumbai, India <i>2020 – 2021</i>